

Yingrui Hou

Data Scientist IA industrielle | Validation, Monitoring & Industrialisation

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SUMMARY

Data scientist titulaire d'un doctorat en physique expérimentale, avec plusieurs années d'expérience dans le développement de workflows Python/C++ reproductibles pour la modélisation statistique, la validation, la calibration et le monitoring de systèmes de mesure complexes.

Mon point fort est de transformer des données réelles bruitées et des modèles imparfaits en solutions fiables et exploitables, en combinant diagnostic d'anomalies, correction des biais, suivi des performances et collaboration avec des équipes techniques multidisciplinaires.

CORE SKILLS

- Data science and AI: Python, pandas, NumPy, scikit-learn, statsmodels, CatBoost / Gradient Boosting
- Validation and monitoring: model validation, calibration, anomaly detection, dataset-shift diagnostics, bias correction, uncertainty-aware analysis
- Industrialisation foundation: Git, CI/CD, Linux, modular and reusable workflows, YAML / Shell / CMake automation
- Delivery and collaboration: technical documentation, structured model outputs, data-quality checks, cross-functional teamwork, knowledge transfer
- Engineering and scientific computing: C/C++, ROOT / RooFit, Geant4, large-scale data-processing workflows

PROFESSIONAL EXPERIENCE

Postdoctoral Researcher | CNRS / CERN / LHCb Collaboration + GRAiNITA Collaboration | 2021-2025

- Built reusable Python/C++ workflows to analyse noisy real-world measurement data, validate model behaviour, and deliver reliable outputs for downstream technical decisions.
- Designed monitoring and diagnostic pipelines to compare expected versus observed behaviour, detect anomalies and distribution shifts, and support root-cause analysis.
- Developed CatBoost-based validation workflows with train/test diagnostics, reproducible model application, feature studies, and dataset-shift mitigation through reweighting.
- Implemented calibration and bias-correction methods that improved model reliability when direct ground truth was incomplete and operating conditions differed between datasets.
- Automated repeatable analysis steps in Linux-based shared codebases using Git, CI/CD practices, YAML, Shell, and CMake, with collaborative review discipline.
- Produced documented analysis methods, structured diagnostics, and interpretable results for multidisciplinary collaborators, clearly communicating model use, assumptions, and limitations.
- Turned validation findings into concrete follow-up actions, including calibration updates and the identification of a software issue that was later corrected downstream.

Doctoral Research | University of Chinese Academy of Sciences | 2016-2021

- Developed simulation-backed statistical workflows combining model fitting, calibration, signal analysis, and bias-aware validation on complex measurement data.

SELECTED OUTCOMES

- Improved reconstructed-energy resolution by about 2x through time-resolved signal analysis and calibration on realistic measurement data.
- Quantified that response non-uniformity contributed less than 1% to system-resolution degradation, helping distinguish genuine system risk from modelling effects.

SELECTED PROJECTS

Test-Data Validation & Root-Cause Analysis

Built a production-style validation workflow for early Run 3 photon reconstruction, focused on finding actionable detector and software issues.

Methods: benchmark-channel checks, MC-data comparison, detector-level diagnostics

Domain Adaptation for ML

Built an ML validation and sample-alignment workflow in a shared analysis environment for noisy selection problems under dataset shift.

Methods: CatBoost training, train-test diagnostics, gradient-boosting reweighting

Model Correction Workflow

Developed simulation-correction tooling that turns measured physics constraints into an event-level resampling workflow.

Methods: Gaudi-package development, event-level correction, framework integration

Time-Series Inference

Built a bias-corrected time-series inference workflow for data affected by selection effects, detector resolution, and sample mismatch.

Methods: acceptance modelling, resolution calibration, custom fit components, joint fitting

Weak-Signal Inference

Built a reusable C++/Python/ROOT workflow for weak-signal extraction in data where background dominates and event-level truth is unavailable.

Methods: likelihood fitting, control-sample validation, production yield workflows

EDUCATION

- Joint Master and PhD in Particle Physics - University of Chinese Academy of Sciences, 2016-2021
- BSc in Applied Physics - China University of Mining and Technology, 2010-2014
- Languages: Chinese: Native speaker, English: C1, French: B1 to B2